Can Data Selection and Preprocessing Improve Image-Text Retrieval?

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What will we build?

We will train a vision–language model from scratch and compare how image selection and input preparation affect retrieval when the training budget is held constant.

Dataset selection

Random selection (baseline)

Object-category balancing

Visual diversity using CLIP

Caption-semantic diversity

Quality-aware diversity

Input preparation

Standard RGB resize

Aspect-ratio-preserving resize

Random resized crop

Color and contrast normalization

RGB + frozen CLIP features

Experiment plan 9 initial runs + up to 5 targeted follow-ups

On what data?

Training · COCO train2017

15,000

unique training images per configuration

One of five captions sampled each epoch

Evaluation · COCO val2017

≈5,000

held-out validation images

Five reference captions per image

Initial design · 9 runs

	P1	P2	P3	P4	P5
D1	B	✓	✓	✓	✓
D2	✓				
D3	✓				
D4	✓				
D5	✓				

D1: random selection · P1: standard RGB · B: baseline

Follow-up

Select up to 5 additional
method pairings based on performance

Compute Estimation

Assumed model ViT-B/16 at 640 × 640 + 12-layer text Transformer

Forward pass	371.03 GFLOPs per image–caption pair
Training multiplier	3 × forward-pass compute
One configuration	15,000 images × 40 epochs
Training plan	9 initial runs + up to 5 follow-ups

Estimated total training compute

≈ (6.01–9.35) × 10^{18} FLOPs

Est. to Train On Single CURC Alpine A100 GPU

How will we know it worked?

Retrieval tasks

Image → text	Rank captions; five valid matches per image
Text → image	Rank images; one correct image per caption
Measure	Recall@1 Recall@5 Recall@10
Compare	Each configuration against random selection with the same image and training-epoch budget